KOH Boon Xiong Ariq

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EDUCATION

Nanyang Technological University, Singapore — Aug 2024 – May 2028
Bachelor of Computing (Computer Science), Turing AI Scholars Programme (TAISP)

- Expected Honours (Distinction), Current CGPA: 4.65/5.00
- **Relevant Coursework:** Artificial Intelligence (A+), Data Structures & Algorithms (A+), Object Oriented Design & Programming (A+), Probability (A+), Digital Logic (A+), Algorithm Design & Analysis (A), Computer Organisation & Architecture (A), Calculus (A)

Victoria Junior College, Singapore — Jan 2020 – Dec 2021
Singapore-Cambridge GCE Advanced Level

- Grade: 88.75/90.00 Rank Points

SCHOLARSHIPS

Turing AI Scholars Programme (TAISP) Scholarship — Aug 2024 – May 2028
Awarded by Nanyang Technological University, Singapore

PUBLICATIONS

QD-LLMs: Quality Diversity Optimization with LLMs for Generative Design Exploration — Jul 2026
First Author, In Proceedings – GECCO 2026 Workshop, San José, Costa Rica

Quality-Diversity LLM for Generative Design — Jun 2026
First Author, In Proceedings – AI4X Accelerate 2026, Singapore

EXPERIENCE

A*STAR Centre for Frontier AI Research (CFAR) — Apr 2026 – Aug 2026
Research Intern – Agentic Scientist Platform

- Building an Agentic Scientist Platform on a multi-agent framework for automated scientific research.
- Implemented using Google ADK, incorporating modern agentic techniques including RAG, tool calling, and multi-agent orchestration.

Alibaba-NTU Global e-Sustainability CorpLab (ANGEL) — Jun 2025 – Dec 2025
Research Intern – Evolutionary Prompt Optimization

- Implemented a black-box automatic prompt optimization (APO) framework for LLM powered chatbot systems.
- Designed and built a multi-agent system built on evolutionary strategies to iteratively refine and optimize chatbot system prompts without access to model parameters.

PROJECTS

NTU Undergraduate Research Experience on Campus (URECA) — Aug 2025 – Present

- Proposed a novel framework for generative design exploration with LLM-based variation operators.
- Demonstrated more than 80% better physical performance in producing domain-aligned geometries against baseline methods, including iterative zero-shot prompting and Covariance Matrix Adaptation MAP-Elites (CMA-ME).

OneStop · iOS App with On-Device AI Assistant — Jan 2026 – Present

- Designed and deployed an all-in-one campus assistant app on the App Store, consolidating timetable management, bus tracking, and campus navigation into a single platform for NTU students.
- Built on-device AI chat powered by Apple Foundation Models (3B parameter model on Apple Neural Engine), with automatic timetable extraction from screenshots and chatbot-integrated timetable queries and class reminders.
- Achieved 50+ downloads on the App Store within 1 week of launch.

AWARDS

NTU Deep Learning Week 2026 · Finalist, Top 10 in Track — Mar 2026

- Built ResNet, an end-to-end post-crisis resource allocation system that analyzes disaster social media posts to predict emergency resource needs and visualize them on an interactive map, as part of a team of 3.
- Combined multimodal deep learning (CLIP ViT-B/32 encoder, ViT damage severity classifier, MLP resource classifier) with DBSCAN spatial clustering, and LLM-powered crisis summaries in a React + Leaflet frontend.

TAISP Hackathon 2025 · 3rd Place — Dec 2025

- Built real-time ASL recognition backend: MediaPipe hand landmark detection → TensorFlow/Keras neural network classifier (A-Z letters) → FastAPI WebSocket server streaming predictions at 30 FPS.
- Implemented GestureBuffer for temporal smoothing, pose analysis feedback, and word-completion detection mode for educational Visual Novel game integration.

SKILLS

Languages: Python, C, C++, Java, Swift, English, Chinese
ML/AI: PyTorch, TensorFlow/Keras, pyribs (QD optimization), OpenAI API, LangChain, MediaPipe
Tools: Git, NumPy, OpenCV, SwiftUI, Node.js/Express